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RESEARCH ARTICLE

Prediction of Cardiovascular Disease Risk From Retinal Vasculature Using a Quantitative Diagnostic Approach With CVD-Net in DR and HR Patients

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ABSTRACT The global incidence of diabetes is increasing significantly each year, posing serious health risks if not diagnosed at an early stage. Elevated blood glucose levels can lead to complications such as Diabetic Retinopathy (DR), Hypertensive Retinopathy (HR), Cardiovascular Diseases (CVD), and renal failure. **Background** The objective of this study is to establish an association between retinopathy and CVD, which is driven by shared pathogenic mechanisms including inflammation, microvascular damage, oxidative stress, and endothelial dysfunction. Diabetes and hypertension substantially increase the likelihood of DR and HR, which in turn elevate the risk of developing CVD. **Objective** This work is to forecast the risk of cardiovascular diseases by analyzing retinal vasculature and associated clinical risk factors in patients with DR and HR. **Methods** The method involves a quantitative diagnostic approach using morphological and physiological attributes of the retinal vascular system, such as the Arteriole-to-Venule ratio (AVR) and the Cup-to-Disc ratio (CDR), which serve as biomarkers. In addition, systemic risk indicators including age, gender, Body Mass Index (BMI), smoking habits, and alcohol consumption are incorporated. Deep Learning (DL) techniques are employed to detect DR, HR, and CVD and to quantify their characteristic features. **Results** The results indicate that early investigations based on the proposed approach can effectively identify patients at high risk, potentially preventing up to 90% of CVD cases when detected during the early stages. **Implications** This study highlight the potential of retinal imaging, when combined with deep learning, to serve as a reliable, non-invasive tool for early cardiovascular risk prediction in diabetic and hypertensive individuals, enabling timely clinical intervention.

INDEX TERMS Arteriole to venule ratio, cardiovascular disease, deep learning, diabetic retinopathy, hypertensive retinopathy, retinal vasculature, optic cup to disc ratio.

I. INTRODUCTION

According to a World Health Organization (WHO) report, cardiovascular disease (CVD) accounts for more than one-third of all deaths globally and is the leading cause of mortality worldwide [1]. Atherosclerosis development is also regarded as a significant risk factor for CVD, alongside other variables such as chronic Diabetes Mellitus (DM), hypertension, and lifestyle choices. The two main causes of death worldwide today are diabetes and hypertension, which also have a significant role in the development of heart attacks, strokes, chronic kidney illnesses, and blindness. Regardless of any risk factors, those with diabetes have an approximately two-fold greater risk of CVD. This is a microvascular effect of diabetes, encompassing conditions such as retinal degeneration, coronary artery disease, and cerebrovascular sickness [2]. More specifically, it tends to result in DR, a chronic condition that impacts visual function. However, additional factors that pertain to one's lifestyle, including smoking, an unhealthy diet, stress, and a lack of physical activity, might also hold significance.

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DR is a serious vision-threatening consequence of diabetes, which stands as the prevailing chronic microvascular complication and remains the leading cause of blindness in working-age persons. It is typically identified through regular screening, often using retinal fundus photographs [3]. On account of resemblances in their risk factors and underlying mechanisms, which encompass inflammatory factors and endothelial impairment, an inherent link between DR and CV events is anticipated. The existence of retinopathy in individuals with diabetes is recognized to correlate with an elevated risk of CVD [4]. Moreover, high blood pressure (hypertension) significantly increases the risk of acquiring several health conditions, like HR and stroke. This damages the delicate blood vessels, especially the walls of arteries. A greater risk of total CVD was linked to mild hypertensive retinopathy in those with unregulated blood pressure.

Most occurrences of CVD are preventable by the management of risk factors, which include both personal factors like smoking habits, blood pressure readings, BMI, and other metabolic variables like blood sugar and lipid levels. Recent research indicates that the metrics of microvascular impairment have connections to CVD, extending beyond the general population to diabetes-specific groups. This implies that utilizing retinal measurements derived from retinal images, which portray widespread microvascular issues, holds the potential to enhance the precision of assessing CVD risk among individuals with DR and HR [5]. In addition to these qualitative indicators of vascular abnormalities, novel markers of microvascular impairment, such as alterations in retinal vascular caliber, can now be quantified from the very same retinal images using computer-assisted software. with unregulated blood pressure.

The prevailing approaches to evaluating the risk of CVD involve examining many parameters obtained from the individual's medical information. Moreover, due to the possibility of visualizing blood vessels non-invasively through retinal fundus images, a range of characteristics within the retina, including vessel caliber, optic cup to disc risk ratio values, microvascular alternations, and vascular fractal dimensions, have the potential to indicate both the overall health of the cardiovascular system and future risks. This study demonstrates the use of DL to assess and extract various cardiovascular risk markers from retinal images [6]. Hence, the study investigated the potential to extract further indicators of CV risk that can be swiftly, affordably, and non-invasively acquired from an automated system environment. Indicators of CV illness, like HR and DR, frequently become evident in the eye.

A. RESEARCH GAPS

Despite growing evidence that retinal microvascular measurements improve CVD risk assessment, existing studies generally:

- Analyse only isolated retinal features (e.g., caliber alone) or rely on semi-automated/manual annotation, which limits scalability.
- Seldom integrate systemic risk indicators (age, BMI, smoking, alcohol consumption) with retinal biomarkers in a single predictive model.
- Lack a specialised DL architecture designed to jointly classify DR, HR, and stratify CVD risk.

B. MOTIVATION

- The rising global burden of diabetes and hypertension, coupled with their strong association with CVD, necessitates the development of early, accurate, and non-invasive diagnostic solutions.
- Retinal vasculature reflects systemic microvascular health, offering a unique opportunity to detect early signs of CVD through retinal imaging—before clinical symptoms appear.
- Existing cardiovascular risk assessment methods are often invasive, costly, or inaccessible in low-resource settings, underscoring the need for an automated, scalable, and affordable screening approach.
- Manual analysis of retinal images is time-consuming and prone to inter-observer variability, creating a demand for robust deep learning models capable of extracting and interpreting retinal biomarkers consistently.
- Integrating retinal biomarkers with demographic and lifestyle risk factors into a single diagnostic framework may significantly improve the accuracy and efficiency of CVD risk prediction, especially in diabetic and hypertensive populations.

C. OBJECTIVES

The timely identification of CVD is crucial for effective clinical intervention, where the integration of DL models can enhance the diagnostic procedure [7]. The objective of this study is to examine the primary advancements in automated cardiovascular disease diagnosis using retinal fundus images. The majority of cardiovascular risk calculators employ a blend of the parameters extracted from the retinal image to detect individuals who are likely to encounter a significant cardiovascular event or mortality related to the heart within a predetermined timeframe, often around five years.

This research provides a framework that utilizes DL models to address the challenges listed below. The main contributions of the suggested approach are:

- A new quantitative diagnostic method was developed that classify the risk of CVD in individuals with hypertension and diabetes. This methodology involves examining the blood vessels in the retina.
- The suggested framework shows that existence of DR and HR is indicative of a heightened risk of CVD.
- To estimate DR, HR, and cardiovascular risk stratification, a new framework is suggested based on the use of cutoff values for AVR Ratios.
- In this paper CDR thresholds are used in predicting the probability of developing DR, HR and CVD.

- This paper proposes a distinct neural network architecture for the prediction of cardiovascular risk factors in retinal photographs.

The following portion of this paper is categorized as: Section II presents a summary of the previous research. Section III provides a detailed explanation of the methodologies and dataset used in the diagnostic model. Section IV provides an explanation of the experimental findings and the results derived from the model. Section V outlines ablation research. The research is concluded in Section VI.

II. LITERATURE REVIEW

Several existing works could be observed and considered for the calculation of AVR and CDR and the primary risk factors of CVD. Cheung et al. [8] have presented DL models to automatically quantify retinal vascular diameter with the aim of predicting CVD risk using the SIVA Deep Learning interface and manual assessments by trained human graders. These algorithms might perform better or be similar to those of professionals in analyzing the connections between retinal artery size measures and risk factors for CVD. It could be enumerated as a measurement of total cholesterol levels, hemoglobin with glycated level, elevated blood pressure, and a person's body BMI.

The work suggested by French et al. [9] investigates CV risk with semi-automated calculators QRISK2 and Mayo Clinic. After undergoing the consistent eye examination, participants had retinal visualization, specifically focusing on the optic disc. Additionally, their systemic blood pressure was measured. The measurement of retinal vascular characteristics, especially the central retinal artery and vein equivalent, as well as AVR, was conducted using software that requires some human input.

Rajan [10] provided an approach to collecting data from retinal tissues, which is then employed for the identification and control of CV ailments including stroke, hypertension, glaucoma, and high blood pressure. The retinal image that has been collected is subjected to filtering and then divided into several segments. Their findings are employed for the classification of arteries and veins utilizing the Support Vector Machine (SVM). The CDR is ascertained by quantifying the dimensions of the optic cup and disc. Artificial Neural Networks (ANN) are employed to identify the presence of CVD and evaluate its parameters.

Taha et al. [11] proposed a technique that has the capacity to enable doctors to automate the process of finding novel insights and disease biomarkers. The aim of this work was to develop an advanced deep learning system capable of properly predicting the gender of people using retinal fundus images. The model addressed was built based on the pre-trained Xception model.

The work proposed by Poplin et al. [7] was a novel approach for utilizing deep learning techniques specifically for analyzing retinal fundus images. The major findings were to foresee several CV-associated variables, including age, gender, and Systolic Blood Pressure (SBP). The incorporation of these risk indicators, which are essential components in several cardiovascular risk calculators, implies that the proposed algorithm possesses the ability to accurately predict cardiovascular risk. The soft attention technique is employed in order to identify the anatomical regions that are linked to risk factors. As a result, a predictive model was built and subsequently trained to anticipate the incidence of CVD during a five-year period.

Previous studies have shown that the majority of approaches used semi-automated software to evaluate the subclinical impairment of retinal microvascular structure shown in the Table 1. All previous works have carried out the classification of the artery and vein and the optic cup-to-disc ratio for predicting HR and DR using this semi-automated software [12]. The major limitations in the earlier works were the necessity of additional software that is used to find the relationships between vascular structures, and additionally, there is a need for clinical assistance for further processing [13]. To overcome these limitations, an innovative approach has been proposed. The objective of this approach is to address the structural and anatomical features of retinal vascular scans that would provide more insights. Therefore, it makes it easier to comprehend the retinal microvascular issues associated with the development and growth of cardiovascular disease.

TABLE 1. Summary of existing literature on retinal image-based approaches for cardiovascular disease risk prediction.

Author	Model	Application	Limittaions
Cheung et al. [8]	DL via SIVA + manual grading	Retinal vascular diameter quantification for CVD risk	Partially manual, dependenc y on expert grading
French C et al. [9]	Semi-automated tools (QRISK2, Mayo Clinic)	CV risk estimation using AVR and systemic BP	Semi-automated approach requiring human input
Rajan S.P. et al. [10]	SVM + ANN models	Classification of arteries/veins, CDR estimation for CVD risk	Requires pre-processing; performanc e varies with image quality
Taha A.M. et al. [11]	Pre-trained Xception DL model	Predicting gender using retinal images as a biomarker	Focused only on gender prediction; indirect CVD relevance
Poplin R. et al. [7]	Soft attention-based DL model	Predicting CV risk factors (age, gender, SBP) from retinal images	Requires large datasets

III. CVD-NET METHODOLOGY

The present research offered an innovative framework for CVD prediction and demonstrated the vascular association

between DR, HR, and CVD using retinal fundus imagery. Recent advancements in retinal imaging technology, as well as its ability to identify DR and HR, provide a significant advantage for risk assessment, so individuals suffering from DR and HR can be evaluated and stratified for CVD directly. Table 2 depicts the proposed procedure for predicting CVD.

A. DATA PREPROCESSING

In the field of medical imaging, Retinal Fundus Imaging (RFI) is characterized by rigorous control measures and limited exposure to ambient noise, resulting in the acquisition of superior -quality inputs. To develop more accurate models and predictions, however, fundus photographs are often preprocessed. The use of gamma correction [14] enables the manipulation of picture brightness.

Data augmentation [15] is a methodology that encompasses the physical altering of a provided sample image via the implementation of diverse image transformations. This process aims to provide additional inputs for the dataset. The purpose of this strategy is to address the problem of overfitting that might occur during the training phase when there is a restricted quantity of input samples accessible. The underlying proposition in this study is that the network may acquire a larger quantity of information by applying augmentations to the original photos.

The computational cost is increased when RFI is directly used in DL models. Therefore, to preserve the aspect ratio of images, they are constantly resized (scaled) before being included in the model [16]. After preprocessing retinal images, the proposed methodologies are used to predict the DR, HR, and risk of CVD.

B. SCENARIO 1: THE APPLICATIONS OF CUTOFF VALUES FOR ARTERIO VENOUS (AVR) RATIOS IN PREDICTING DR, HR AND CARDIOVASCULAR RISK STRATIFICATION

The use of fundus photography, in tandem with retinal vascular analysis for the purpose of calculating the AVR facilitates the non-invasive evaluation of the condition of arteriolar walls. Arterial stenosis [17] is an indicator of increased arterial burden caused by DR, HR complications and many CVD Risk factors and it has the capacity to predict the probability of stroke [18], [19].

Multiple comprehensive population-based studies, such as the Atherosclerosis Risk in Communities (ARIC) project [20] and the Beaver Dam study [21], have provided empirical support for a correlation between the attributes of the AVR along with DR [22]. The examinations conducted by the researchers unveiled a decrease in the diameter of the aortic valve annulus, which was determined to be a possible indicator for the advancement of DR. In addition to occurrence of DR, there have been observations of other microvascular complications, such as HR. Within the realm of hypertension, there is available evidence indicating a relation between hypertension and the constriction of retinal arterioles. As to the Keith-Wagener-Barker [23] classification, an AVR value below 0.66 signifies the quantification of retinal vascular diameter is frequently denoted by the AVR.

TABLE 2. Pseudocode of CVD-Net model.

ALGORITHM 1:Pseudocode of CVD-Net Model.

Input: High-resolution retinal fundus image (2896 × 1944 pixels)
Patient demographic and lifestyle data
Output: Cardiovascular risk prediction and referral decision

- *Preprocess the input fundus image*
  - *Normalize intensity, enhance contrast, and remove noise*
- *Extract retinal vascular features*
  - *Calculate Arteriole-to-Venule Ratio (AVR)*
  - *Calculate Cup-to-Disc Ratio (CDR)*
- *Evaluate retinopathy status*
  - *If AVR < 0.6 →*
    - *If diabetic → Label as Diabetic Retinopathy*
    - *Else → Label as Hypertensive Retinopathy*
  - *Else → Label as Healthy*
  - *If CDR > 0.3 →*
    - *If diabetic → Update label to Diabetic Retinopathy*
    - *Else → Update label to Hypertensive Retinopathy*
- *Collect additional biomarkers:*
  - *Age, Gender, Body Mass Index (BMI), Smoking Status, Alcohol Consumption*
- *Cardiovascular Risk Prediction*
  - *If (AVR < 0.6) AND (CDR > 0.3) AND (biomarker condition is True):*
    - *Predict high cardiovascular risk*
    - *Recommend referral to cardiologist*
  - *Else:*
    - *Predict healthy retina*
- *Return predicted CVD risk status and referral suggestion*
- *Stop*

The assessment of structural abnormalities in the retinal microvasculature has been made possible by recent developments in imaging technology and automated visual categorization approaches. According to previous findings shows correlation between specific changes in the retina and the existence of CV risk factors. The research investigation pertaining to Beaver Dam study demonstrated a noteworthy association between decreased arteriolar tortuosity and impaired arteriolar bifurcation, and the presence of mortality due to heart disease. The ARIC study reported on a partially automated approach for evaluating vessel calibres in fundus pictures. The research conducted in this paper discussed association between AVR and generalized arteriolar constriction, their potential as predictive factors for incident cardiovascular diseases. These findings were observed to be significant predictors of cardiovascular diseases, regardless of the occurrence of other established cardiovascular risk factors.

The purpose of this study was to examine the possibility of AVR as a reliable long-term measure for assessing the effectiveness of treatment in patients with HR, DR, with

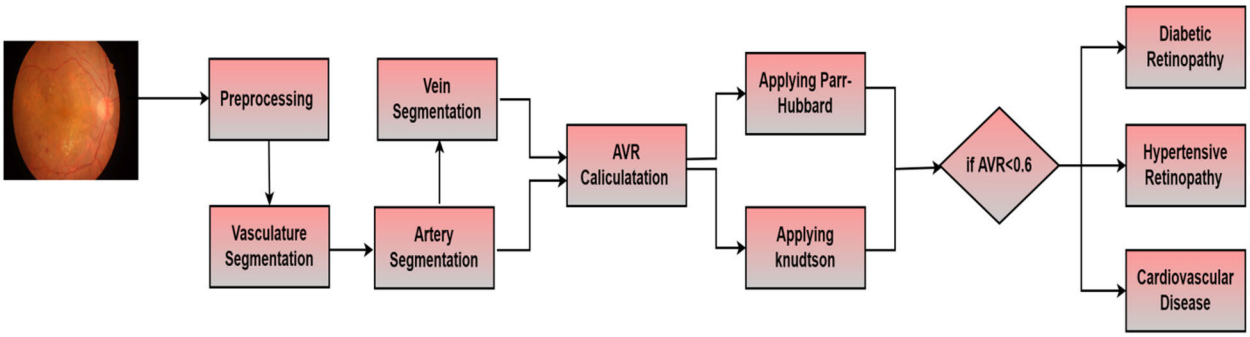


FIGURE 1. Workflow for calculation of AVR values.

the goal of preventing additional cardiovascular events. Fig.1 depicts the calculation of AVR ratio.

1) VESSEL SEGMENTATION AND ARTERY/VEIN CLASSIFICATION

In this research to automatically indentify AVR calculation, the Process of Parallel Segmenting and Classifying Retinal Arteries and Veins (PSCAV) involves the process of distinguishing blood vessels within images [24]. In this work, we use fully convolutional neural networks (FCNNs) to perform an integrated analysis of vascular structures. The FCNNs are trained to provide pixel-wise image segmentations. To achieve efficient segmentation and labelling of vasculature, we use a decomposition approach to address the PSCAV issue. This approach involves dividing the problem into three distinct tasks: segmentation of arteries, segmentation of veins, and segmentation of vessels.

There are several distinguishing characteristics that may be utilized to distinguish arteries from veins. Arteries have a higher level of brightness compared to veins, while also possessing a narrower diameter in relation to adjacent veins. The diameter of the central reflex is comparatively smaller in veins and relatively bigger in arteries. Typically, before branching, arteries and veins alternate close to OD. In many cases, the mentioned features are sufficient to appropriately classify a blood vessel as a vein or artery. Thus, the blood vasculature is used to segment the arteries and veins while using the characteristics. The multi-segmentation approach is in accordance with the intrinsic duality of the vascular network. This duality refers to separate pathways of arteries and veins as they cross the retina, frequently overlapping at different spots. Furthermore, in accordance with this methodology, the neural network immediately generates three distinct segmentation maps for the continuous vascular tree, including arteries, veins, and all other vessels.

In the present scenario, the output of the network comprises three distinct and autonomous channels. Each channel within the dataset encompasses predicted probability map that corresponds to a certain anatomical feature, including arteries, veins, and vessels. The PSCAV is performed utilizing both approaches, using U-Net architecture [25], a fully convolutional neural architecture. The architectural framework was first proposed for the aim of segmenting biological images, and has shown considerable effectiveness in generating favorable results. Additionally, U-Net has been effectively used for classifying veins in arteries and segmenting the vasculature.

2) AVR CALCULATION

The standard way of measuring the retinal arterioles and venules diameter is represented as Central Retinal Arteriolar Equivalent (CRAE) and Central Retinal Venular Equivalent (CRVE), respectively. For the six largest arterioles and venules, these measurements were collected. The AVR is determined by assessing the diameters of blood vessels in two specific regions. The first region, known as zone B, is located 0.5-to-1.0-disc diameters away from the margin of the disc, and includes the ($CRAE_B$, $CRVE_B$). The second region, referred to as zone C, is situated 0.5-to-2.0-disc diameters at a distance from the disc margin, and encompasses ($CRAE_C$, $CRVE_C$) as shown in the Eq. We have fallowed a Parr-Hubbard and Knudtson revised formulas [26] to calculate the two equivalents and represented in (1), (2) and (3),(4). The ROI is evenly sampled to create six distinct areas for separate zones for conducting AVR calculations as shown in (5). To calculate the regional AVR's the six biggest arteries and the six largest veins are determined for each area, and the CRAEs and CRVEs are acquired.

$$CRAE = \sqrt{0.8.Y^2 + 1.01.Z^2 - 0.2.Y.Z - 10.73} \quad (1)$$

$$CRVE = \sqrt{0.72.Y^2 + 0.91.Z^2 + 450.02} \quad (2)$$

$$CRAE = 0.88\sqrt{Y^2+Z^2} \quad (3)$$

$$CRVE = 0.95\sqrt{Y^2+Z^2} \quad (4)$$

where Y is the median of the list of "Arteriole" values, and Z is the value that comes immediately before it. Median "Venule" is Y, and the value preceding Y in the list is Z. The average value of (AVR) may be determined by dividing (1) by (2) and (3) by (4).

$$AVR = \frac{CRAE}{CRVE} \quad (5)$$

The assessment of variations in vessel diameters, known as AVR, has substantial significance in the confirmation and

evaluation of HR and DR [27], [28]. The recommended technique for the INSPIRE dataset has been shown to have an average value over 0.6 in normal circumstances, but falling below 0.6 in instances involving patients diagnosed with DR, HR and CVD illnesses.

C. SCENARIO 2: THE DETERMINATION OF THE OPTIC CUP TO DISC (CDR) THAT RATIO IS A CRUCIAL STEP IN PREDICTING THE LIKELIHOOD OF DR, HR AND CVD

The accurate determination of shape and precise localization of the retinal optic disc has considerable significance in the diagnostic process of retinal disorders, included such as DR, HR, CVD, macular degeneration, and glaucoma [29]. Early diagnosis of retinal disorders has the potential to mitigate vision loss in the affected eye. The quantification of the Cup to Disc Ratio involves the assessment of the dimensions of the optic disc and cup subsequent to their identification [30] as show in the Fig.2. The use of CDR is employed in the prognosis of cardiovascular disease. The typical range for the CDR in the retina of the eye is 0.3. If the CDR ratio deviates from the threshold of 0.3, it signifies an aberrant situation suggestive of the existence of various renal disorders including DR, HR, CVD.

1) OPTIC DISC DETECTION

Optic disc (OD) often manifests as a luminous circular structure inside a retinal view. The location of OD can be identified by selecting the region with the maximum local variance due to its much greater intensity compared to the surrounding retinal the background and the presence of many blood veins originating at OD central region. The procedure of automated item identification border detection entails the use of a snake active contour technique. The snake algorithm [31] has been identified as a very beneficial and efficient method for extracting object boundaries with diverse and arbitrary shapes. The is accomplished by the iterative reduction of an energy function. The optic disc, characterized by its round shape, is surrounded by a set of points, represented as p(s), which are arranged in a circular pattern around the center point. The energy of a snake is determined by (6).

$$E_{snake} = \int_0^1 [E_{int}(p(s)) + E_{image}(p(s))]ds \tag{6}$$

where E_{int} is the snake points' internal energy, a measure of the curve's smoothness and continuity, Eimage represents the image's derived external forces, which may take the form of lines, edges, and terminations. The internal energy is shown in (7)

$$E_{int} = \propto \left|\frac{dp_i}{ds}\right|^2 + \beta \left|\frac{d^2p_i}{ds^2}\right|^2 \tag{7}$$

where an approximation to the first-order differential is in Eq (8)

$$\left|\frac{dp_s}{ds}\right|^2 = |AverageDistance - \|p_s - p_{s-1}\||\tag{8}$$

and the second-order differential can be estimated as Eq (9)

$$\left|\frac{d^2p_s}{ds^2}\right|^2 = |p_{s-1} + 2p_s + p_{s+1}|^2 \tag{9}$$

The energy, given in (8), is a measurement of contour continuity. The contour smoothness is quantified by the value of (9) here. During each iteration, the snake points are adjusted in a manner that minimizes the combined internal and exterior energy as described in (6).

The OD area often has a circular anatomy, characterized by its smooth and uniform outlines and fragmented into many sub-regions due to vascular system. Hence, we have the following attentions for the snake method. To minimize the impact of blood vessel dilation within OD, we increased the weights assigned to smoothness in (7).

In this study, 200*180 pixels, OD region, is obtained from each retinal image using a identification of snake points, where the local contrast and local variance are calculated. The local contrast may serve as a reliable measure for the size of OD owing to the high intensity of brightness of OD. According to the local contrast values, we can set the initial size of OD. The local variance, on the other hand, may reveal the intensity variation of OD associated with its blood-vessel and minor structures.

2) OPTIC CUP SEGMENTATION

After OD is identified, in order to distinguish the optic cup from the backdrop, adaptive threshold T is used. Specifically, if the pixel intensity is greater than T, the intensity is converted to 1. Moreover, if the pixel intensity is equal to or below the threshold value T, then the pixel intensity is set to 0. Equation (10) elucidates the operational mechanism of the threshold T.

$$h(x, y) = \begin{cases} 1, & if\ h(x, y) > T \\ 0 & if\ h(x, y) \leq T \end{cases} \tag{10}$$

T is determined in the following manner (11)

$$T = (0.5 * m) + (2 * \sigma_G) + (2 * \sigma_{GI}) + \mu_{GI} \tag{11}$$

where m represents the size of Gaussian window, σ_G denotes standard deviation of Gaussian window, σ_{GI} is the pre-process result standard deviation and μ_{GI} is the average intensity of pre-process result.

The standard deviation is denoted by σ, is calculated by using (12).

$$\sigma = \sqrt{\frac{1}{M * N}} \sum_{i=1}^{M} \sum_{j=1}^{N} (A(i, j) - A\hat{A}^2) \tag{12}$$

A represents result of the preprocessed image, while M, N denotes the columns and rows and A ^is the Average mean.

A set of morphological operation is executed once the picture has been segmented. There are two stages to the morphological operation: dilation and erosion. The dilation operation results in an increase the size by including pixel values to its boundary as shown in (13) fills the internal

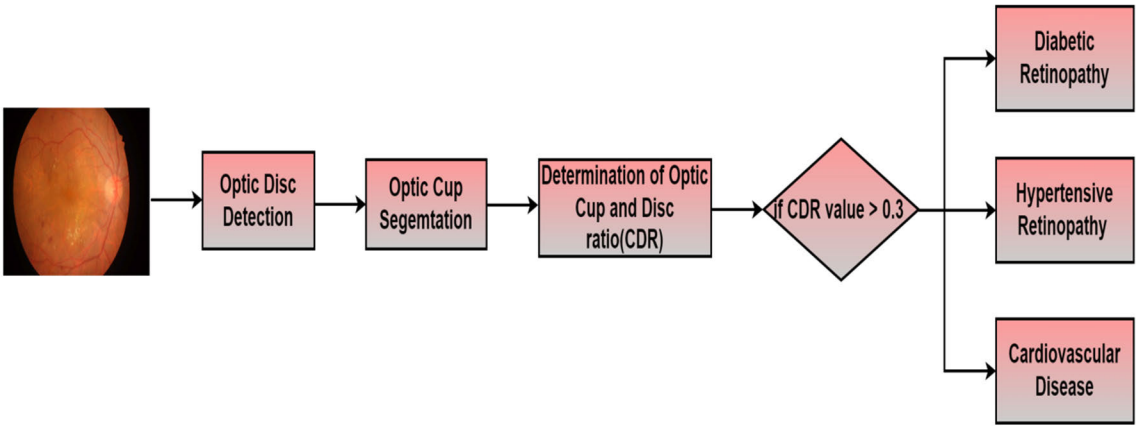


FIGURE 2. Workflow outline for calculating CDR as a predictive measure for DR, HR, CVD.

spaces and lighten the blood vessels by increasing in size the OD influences the CDR ratio. The erosion process that follows dilatation is carried out with the same sized and shaped structural parts. Erosion in (14) is done to contrast the object's boundaries. The end result is an image smoothing with no blood vessels visible.

The C by D is dilated given by

$$C \oplus D = \cup_{d \in D} C_d \tag{13}$$

In addition, C by D is given by

$$CD \cap_{d \in D} \; C_{-d} \tag{14}$$

3) DETERMINATION OF CDR

The ratio between area of a cup to the area of a disc is known as the "Cup to Disc Ratio," and it calculates in three dimensions Horizontal, Vertical, Area. A ratio equal to or less than 0.3 is indicative of a healthy eye. When the ratio above a threshold of 0.3, it is considered indicative of an abnormal eye condition.

The Horizontal CDR ratio H_{CDR} is a metric used to determine the horizontal ratio. It is computed using the equation shown below as (15).

$$H_{CDR} = {}^{H_{CUP}}/_{H_{DISC}} \tag{15}$$

H_{CUP}, H_{DISC} represents horizontal distance separating the optic cup and the optic disc.

$V_{CDR}$is the formula used to ascertain the vertical ratio, and it is given by (16).

$$V_{CDR} = {}^{V_{CUP}}/_{V_{DISC}} \tag{16}$$

V_{CUP}, V_{DISC} is the vertical length of the optic cup and the optic disk.

D. SCENARIO 3: APPLICATION OF DEEP LEARNING METHODS FOR PREDICTION OF RETINAL PHOTOGRAPHIC CARDIOVASCULAR RISK FACTORS

The conventional method for estimating risk of developing CVD necessitates the use of data obtained from blood tests. The main goal of this project is to create and implement a computerized system that can precisely detect cardiovascular risk factors by analyzing fundus retinal images. This work used data obtained from TREND datasets, and applied a ResNet-50 Deep Learning model [32] to forecast the cumulative incidence of severe CVD events over a period of 5- years.

There are certain health disorders that serve as indications of cardiovascular risk, including both metabolic factors and human characteristics such as age and lifestyle. Biomarkers like Age, Gender, Smoking status, Alcohol status, Body Mass Index (BMI) are often regarded as a surrogate measure and play a crucial role in the diagnosis of CVD. The planned architecture consists of two stages as illustrated in Fig.3.

The first phase involves the use of Grad-CAM mechanism [33] was implemented to ascertain the anatomical regions that the algorithm potentially relied in order to generate its predictions. The provided visual representation shown in Fig.3 presents a particular instance of an individual retinal fundus picture, along with corresponding attention maps, often referred to as saliency maps, for each prediction to assess individual CVD risk variables as outcomes.

In the Second phase, Transfer Learning was adopted, using the ResNet-50 model for the purpose of predicting biomarkers. The features derived from the attention maps is used as an input for the network model for the purpose of estimating the features.

Subsequently, the neural network undergoes training to provide many predictions concurrently, including age, gender, smoking status, alcohol status, and BMI. The network weights underwent training utilizing a distributed stocastic gradient-descent implementation for optimization method. In order to enhance the efficiency of the training process, techniques such as batch normalization and pre-initialization using weights obtained from a network trained on the ImageNet dataset were used.

The trained models displayed a tendency to emphasize the perivascular surroundings when generating predictions for CVD biomarkers. The predictive model used to assess

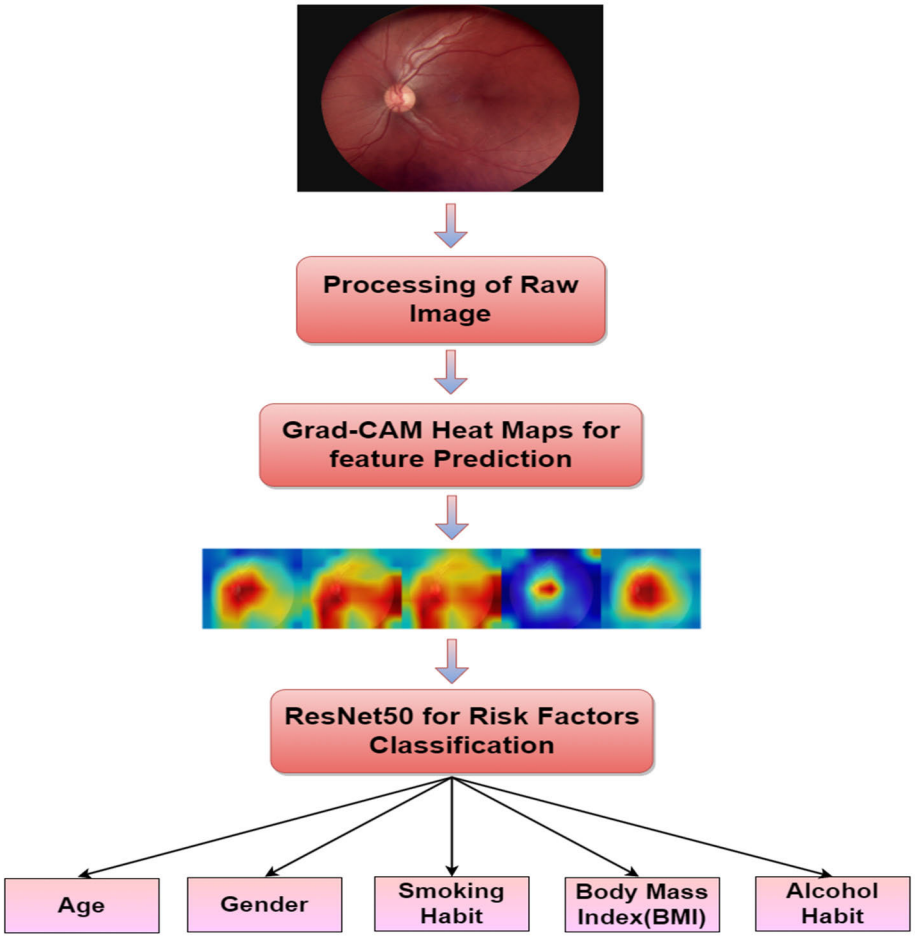


FIGURE 3. The suggested workflow paradigm incorporates the usage of ResNet 50 deep networks for the acquisition of retinal biomarkers.

risk variables, including age, gender, smoking, alcohol consumption, and body mass index, have shown encouraging outcomes by effectively emphasizing the significance of blood vessels.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

In this section, we provide datasets and the objective and quantitative investigation of various studies. These investigations include recommendations for selecting the most appropriate retinal samples, determining the cutoff values for AVR, estimating the CDR of the optic nerve, and collecting retinal biomarkers. All these three investigations are utilized to predict CVD in individuals who have already been suffering with DR and HR. For conducting these experiments, high-resolution retinal fundus images with a size of 2896 $\times$ 1944 pixels, stored in PNG format, were used as input. The resolution ensures the preservation of fine retinal structures required for accurate vascular and anatomical feature analysis.The three experiments in this section provide an in-depth explanation of the computational methodologies used to conduct these studies.

A. DATASETS

In this study, three datasets were utilized. The first dataset, known as the TREND dataset [34], was acquired through an observational study done at the University of Montenegro. The research was carried out between February 18 and March 11, 2020. Each participant in the study granted informed consent and undertook a series of health examinations and questionnaires. The survey included inquiries related to many factors, including age, gender, body weight, height, past medical records, concurrent medicines, and social background (possible use of alcohol, cigarettes, and substances). People were considered to be alcohol users if they consumed any amount of alcoholic beverages, whether every day or occasionally. Conversely, individuals who reported refraining from alcohol intake were categorized as alcohol non-users. Individuals were classified as current smokers if they reported a regular or occasional cigarette smoking habit. Non-smokers were individuals who had never engaged in smoking or had formerly been smokers but had now ceased the habit. The Body Mass Index (BMI) was determined by using weight and height measurements, as per the equation BMI = weight

(kg) / height2 (m2). Individuals with a body mass index (BMI) falling between the range of 18.5 and 24.9 kg/m2 were designated as having a normal weight, while those with a BMI between 25 and 29.9 kg/m2 were categorized as being overweight. Individuals with a body mass index (BMI) greater than 30 kg/m2 were classified as obese. All the individuals included in the study underwent retinal fundus and Optical Coherence Tomography (OCT) imaging using a non-mydriatic hand-held portable camera. To facilitate the creation of our models, we allocated 80% of the TREND dataset to the development dataset, while the remaining 20% was used to evaluate the quality of our models. Low-quality images were removed from the training and validation sets.

The INSPIRE-AVR database [35] comprises a collection of forty high-resolution images, with each image having a resolution of 2392 by 2048 pixels, with a specific focus on capturing the optic disc. These photographs were taken using a Zeiss Fundus camera. The dataset contains two AVR measures calculated by two eye specialists using partially automated computer software named IVAN.

The DRISHTI-GS dataset [36] consists of segmented images of OD and OC, along with their equivalent ground truth images. There are a total of 101 photographs in the collection. The assortment comprises fundus photographs that primarily focus on the ocular disk (OD) and have a field of view spanning 30 degrees. The photographs possess a resolution of 2896∗ 1944 pixels and are stored in the PNG format. The aforementioned dataset is readily available to the general public.

B. EXPREMENT-1: CALCULATING ARTERIOVENOUS RATIO CUT-OFFS FOR PREDICTING DR, HR AND CVD

In experimental settings, INSPIRE-AVR datasets are often partitioned as training set and the testing set. For training purposes, a random sample of 65% of the data is given, while the remaining 35% has been set for testing. The experiment was conducted in a repetitive manner, with the process being performed a total of ten times. The average outcomes of these repetitions are reported in this analysis. The assessment of the suggested model was performed for predicting DR, HR and CVD by calculating AVR Score and their cut-off values represent the presence of diseases or not.

In AVR calculation, segmentation of blood vessels, next Artery vein classification and their diameters plays an important role. Fig.4 displays randomly selected image from the database, which have been annotated to highlight the presence of vascular structures such as arterioles and venules. Additionally, Fig.4b, c, d showcases the mapping of the artery and vein network onto the original fundus image. In addition, the widths of artery fragments from the six largest arteries and veins form Zone B, C were chosen for the purpose of calculating the arteriovenous width ratio in Fig.4e, f, g, h, i, j. In order to determine the arteriovenous ratio, we used the Parr-Hubbard and Knudtson equations. The values of CRAE, CRVE, and AVR for the input are shown in Table 3. In previous studies, these equations have been used to enhance the precision of quantifying AVR.

In this experiment, in order to estimate the overall performance of the proposed model with the assessment of the AVR ratio involves the use of the Root Mean Square Error (RMSE), which is used to compare the categorized AVR values with the ground truths that are provided in the dataset.

TABLE 3. Measured vascular features of six largest arteries and veins in zone B and zone C.

Methods	Calculations	CRAE	CRVE	AVR
Parr-Hubbard	ZONE B	232.855	352.070	0.661
	ZONE C	258.981	398.035	0.650
Knudtson	ZONE B	202.822	202.822	0.524
	ZONE C	228.923	441.405	0.518

TABLE 4. RMSE values for the calculations of Zone B & C for Parr-Hubbard & Knudtson.

Methods	Calculations	RMSE
Parr-Hubbard	ZONE B	0.92
	ZONE C	0.89
Knudtson	ZONE B	0.75
	ZONE C	0.72

The RMSE is used to evaluate the estimation of AVR for all zones as depicted in the Table 4. The mathematical equation for the RMSE is shown in (17).

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n} (AVR_{Actual} - AVR_{Predicted})^2}{n}} \tag{17}$$

where $AVR_{Predicted}$ is value of automated AVR calculation method and AVR_{Actual} is ground truths of two ophthalmologists in datasets.

C. EXPREMENT-2: OPTICAL CUP-TO-DISC RATIO CALCULUS FOR CARDIOVASCULAR DISEASE PREDICTION

The suggested model utilizes the IDRID database for the purpose of collecting retinal photographs. The IDRID database is utilized for the purpose of identifying and randomly selecting an image, subsequently undergoing the implementation of pre-processing techniques and subsequent segmentation. The calculation of CDR has been proposed as a potential predictor of DR, HR, CVD. In the process of calculating CDR, the initial step involves the identification of OD, followed by the segmentation of the optic cup. The "Cup to Disc Ratio" is determined by estimating the ratio of the area of a cup to the area of a disc. This ratio can be calculated in three dimensions: horizontal, vertical, and area. A ratio that is equal to or less than 0.3 is suggestive of ocular health. When the ratio above a threshold of 0.3, it is seen to be symptomatic of an aberrant ocular condition. Fig.5 illustrates the sequential process of localizing and segmenting key anatomical structures of the retina for the computation of CDR, a critical

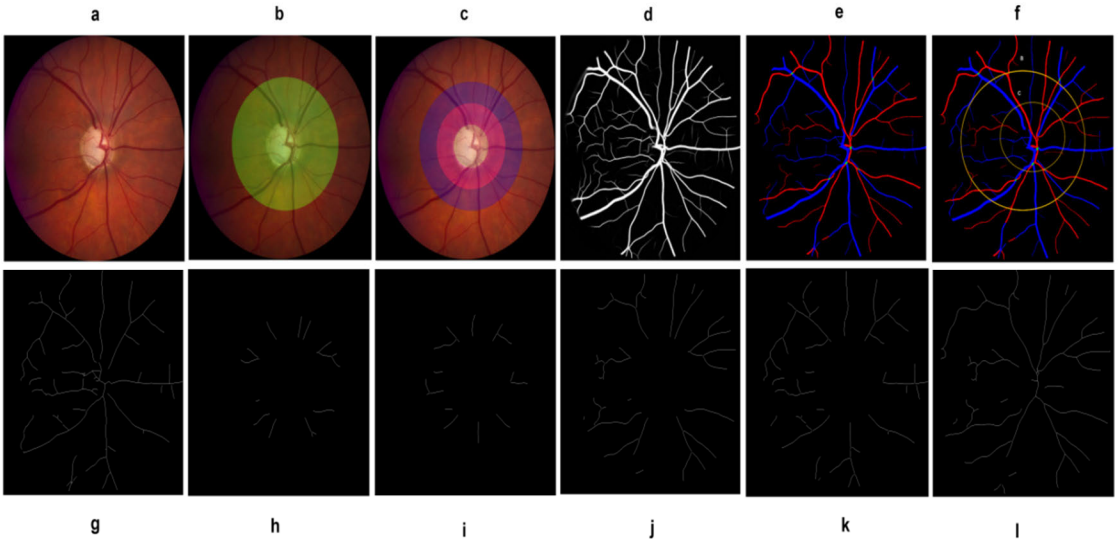


FIGURE 4. Raw retinal image, (b) vascular network after annotation of (a), (c) artery network (red) and vein network (blue), (d) blood vessels were categorized into two concentric zones zone B and C and (e), (f) artery and vein network segmentation (g), (h) zone B artery and vein network, (i), (j) zone c artery and vein network.

biomarker for cardiovascular and ocular disease assessment. Fig5(a) shows the original retinal fundus image. Fig5(b) demonstrates successful detection of the optic disc region, which serves as the foundation for further analysis. Fig5(c), the optic cup is accurately segmented within the disc area, enabling precise measurement. Finally, Fig5 (d) overlays both optic disc and cup boundaries, validating the system's capability to distinguish between the two regions clearly. The quality of segmentation directly impacts the reliability of CDR values, which are essential for identifying hypertensive and diabetic retinopathy–associated cardiovascular risk.

TABLE 5. Performance assessment of normal and cardiac diseased retina.

Raw Image	CDR	Accuracy
Normal	0.296	86.2
Diseased	0.478	90.1

Table 5 represents the some obtained CDR values of the retinal images.The performance analysis of the method is validated by the ground truth obtained of the IDRID Dataset. upon comparing our model with the prior work, the results depicted in the Fig.6 reveal significantly superior accuracy achieved by our suggested model, surpassing the outcomes of the comparative architectures.Table 6 displays the outcomes of comparison between the proposed model and the models utilized in [10]. It can be shown from the performance assessment carried out in this study that the technique that is being suggested is significantly more effective than the models that are already present in the relevant research. We similarly acquired an average score of 0.97 on ACC, but our accuracy score was better than the existing work. Finally, potential of the proposed approach system here easily outperforms the approaches [10].

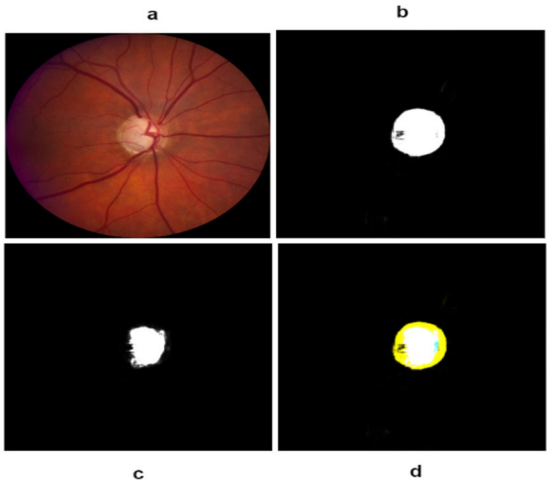


FIGURE 5. (a) Raw image, (b) OD detection, (c) Optic cup segmentation, (d) Optic cup and disc detection.

TABLE 6. Comparison with the existing models.

Author	Model	Accuracy
[10]	Support Vector Machine(SVM)	86.7
Proposed	CDR Method	90.1

D. EXPREMENT-3: PREDICTION OF CVD RISK FACTOR FROM RETINAL FUNDUS IMAGING

The proposed methodology is subjected to a quantitative evaluation for the purpose of assessing variety of Cardiovascular risk factors including Age, Gender, BMI, Alcohol and Smoking from retinal images. The method underwent testing using a sample size of 82 retinal pictures derived from

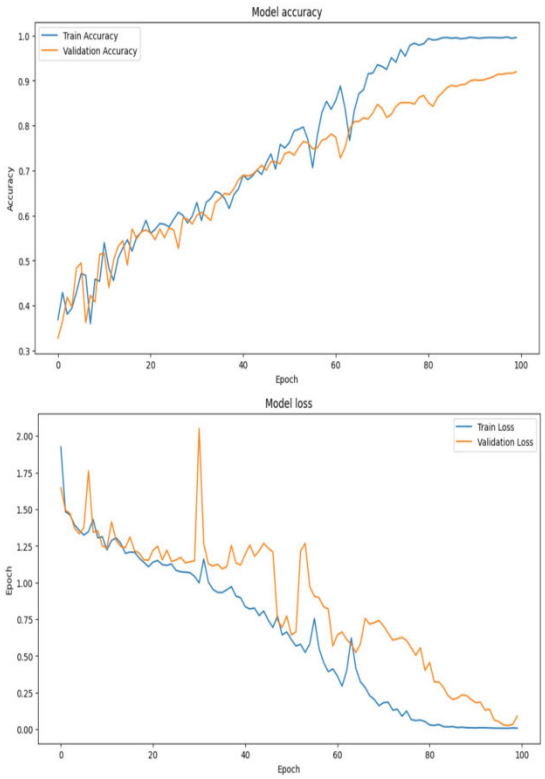


FIGURE 6. Accuracy and Loss or the prediction of CDR ratio.

the TREND dataset. The mean continuous values for Age were 40 years and Participants with a BMI between 18.5 and 24.9 kg/m2 were considered to have normal (healthy) weight, those between 25 and 29.9 kg/m2 overweight, and those over 30 kg/m2 were considered obese. up to 15-30 cigarettes per day are used to predict the risk factors.

These categorical percent values for Age, BMI, Gender, Alcohol use, Smoking status are utilized in the present study showed that the participants who have overweight and obese BMI have Gender, use of alcohol up to drink per day and 3 drinks per week, and smoking habit Initially, we employed Grad-CAM Methods to discern the anatomical locations that the program potentially utilized for its predictions. Fig.7 displays a typical instance of a retinal images together along with accompanying attention maps for each prediction. Remarkably, the algorithms learned to anticipate risk factors, such as Age, Gender, BMI, Alcohol and Smoking, effectively emphasized the blood vessels. Models indicate that age is likely to emphasize the perivascular surrounds. Regarding additional predictions, about Gender and Alcohol status, the attention masks been utilized to emphasize the impacted blood vessels and outline the circular perimeter of the image. The models used for predicting BMI and Smoking mostly focused on OD, blood vessels, and macula.

Subsequently, we assessed the efficacy of our algorithms in anticipating a various of CV risk variables based on images of fundus. Due to the lack of a well-defined benchmark for predicting these characteristics from retinal photographs we rely on the average value as the basis for making continuous assumptions, such as age.

There is a significant association between the expected age and real age, indicating a linear relationship Fig.7. Although the number of events was limited, our model generated an area under the receiver operating characteristic curve (AUC) for every individual risk variable and the mean AUC of 0.91 using only retinal fundus images represented in Fig.8 and in Table 7.

This AUC is comparable to the AUC of 0.72 obtained by existing works Predicting five-year of CVD using various input variables. The information is presented in Table 8. In addition, AUCs were calculated by considering only individual risk variables. Unsurprisingly, the combination of these risk variables was more effective in predicting CVD compared to individual risk factors alone. Furthermore,

TABLE 7. Predicting five-year CVD using various Retinal fundus risk factors.

Factors or model employed for risk prediction	AUC
Age	0.93
Gender	0.89
BMI	0.95
Smoking	0.90
Alcoholic	0.88
Age+Gender+BMI+ Smoking+Alcohol	0.91

TABLE 8. Comparison of CVD Risk factors AUC with previous studies.

Author	Model	Risk Factors Used for Prediction	Performance
Taha A.M et al [11]	Xception	Gender	AUC-0.72
Poplin, R.,et al [7]	Inception-v3	Age, Gender, and SBP.	AUC-0.70
Proposed	ResNet-50 with Grad CAM	BMI,Smoker,Alcohol, Age,Gender	AUC-0.91

TABLE 9. Comparison of Approaches calculating avr values, with and without PSCAV.

	PSCAV	RMSE
With	ZONE B	0.75
	ZONE C	0.72
Without	ZONE B	0.64
	ZONE C	0.62

TABLE 10. Comparison of methodologies for determining CDR values, utilizing with and without Snake Algorithm.

Model	Accuracy
With Snake Algorithm	90.1
Without Snake Algorithm	81.2

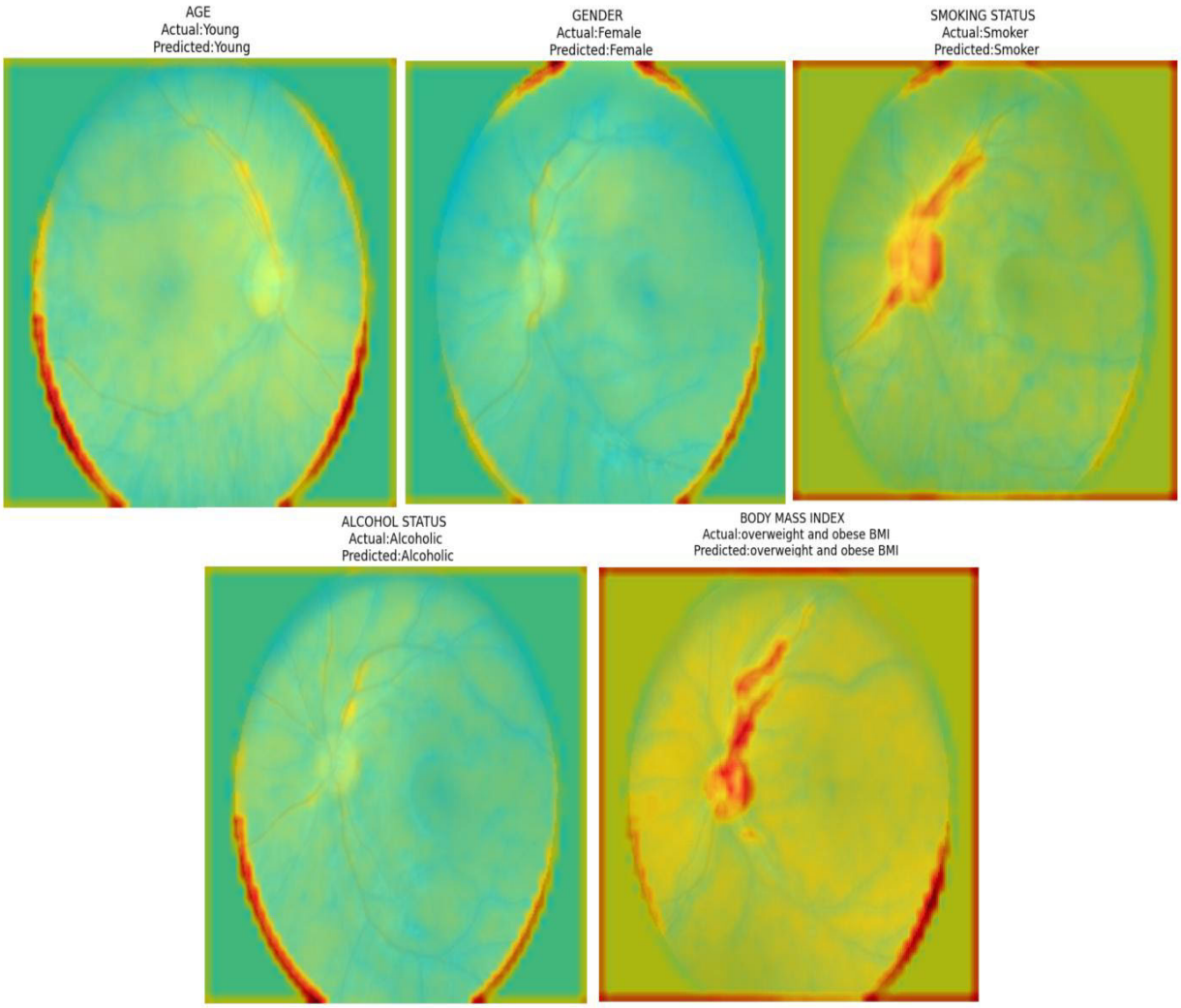


FIGURE 7. Screening of CVD risk factors with the GRAD-CAM model.

metrics of accuracy, precision, recall, F1-score, and confusion are assessed according to diagram presented in Fig.9.

A confusion matrix provides a comprehensive evaluation of a model's effectiveness by comparing the actual identities with projected labels. The evaluation of our model resulted in a confusion matrix for the single-label classification approach, specifically for two classes and all risk factors.Based on the observations that retinal pictures alone were shown to be capable of predicting different cardiovascular risk variables of various degree, it was hypothesized that there may be a direct correlation between these images and cardiovascular events. purpose of forecasting incidence of cardiovascular events during a five-year period.

V. ABLATION STUDY

The Arteriovenous Ratio cut-offs were calculated using retinal pictures to predict CVD, DR, HR. Our study focused on investigating the effects of parallel segmenting and classifying retinal arteries and veins (PSCAV). This method involves the critical task of identifying blood vessels in images, allowing for accurate differentiation between arteries and veins. we achieved the adequate CRAVE and CRVE values for calculation of AVR cut -off-values when compared to the actual values present in the dataset. AVR Ratio achieved lowest RMSE when the PSCAV model was used, compared to the RMSE values obtained without PSCAV as sown in Table 9. The PSCAV model is particularly useful for improving vascular network segmentation. We measured the RMSE values and used them in our comparison study.

We conducted a comparative assessment of CDR ratio with and without snake algorithm in accurate determination of shape and precise localization of OD in retinal fundus pictures has considerable significance in the diagnostic process of retinal disorders, including but not limited to DR,

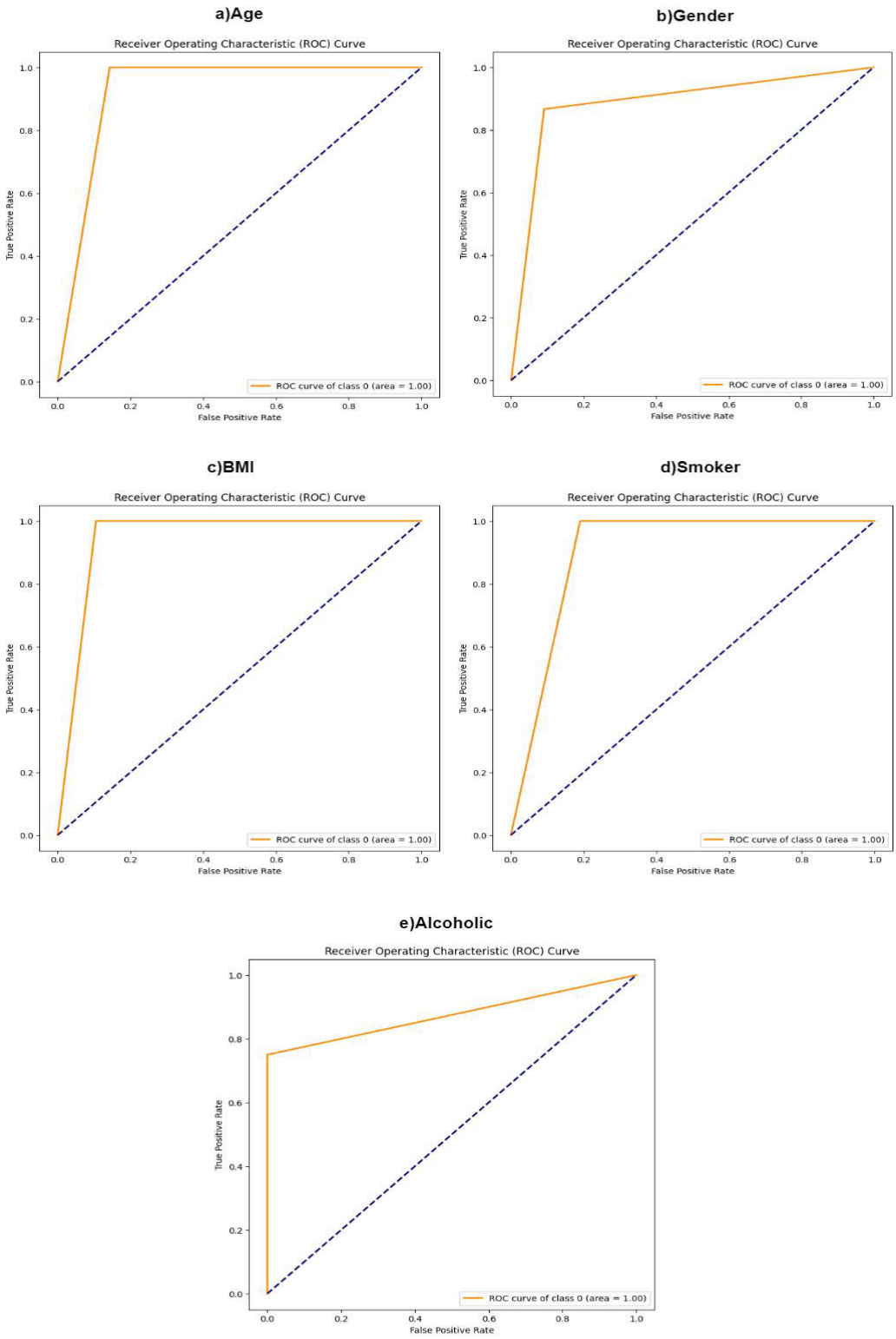
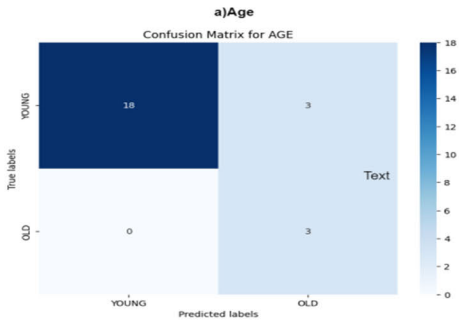


FIGURE 8. AUC representation for a) Age b) Gender c) BMI d) Smoking e) Alcohol Risk factors.

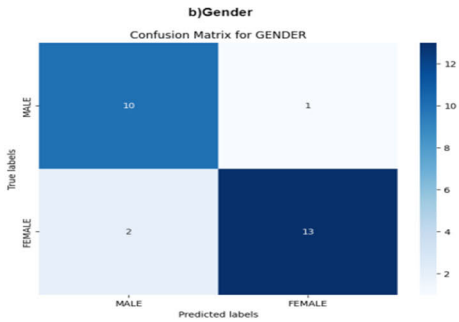
HR, CVD. The findings of this comparison are shown in Table 10. This technique is used to minimize the impact of blood vessels on OD, we increased the weights assigned to smoothness of the curves of vasculature. So, without the incident of snake model the optic disc localization does not achieve high accuracy.



Classification Report:

	precision	recall	f1-score	support
0	1.00	0.86	0.92	21
1	0.50	1.00	0.67	3
accuracy			0.88	24
macro avg	0.75	0.93	0.79	24
weighted avg	0.94	0.88	0.89	24

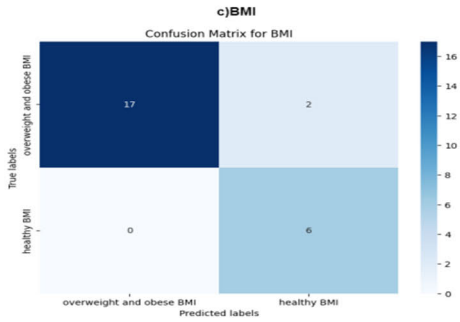
Class 0: Accuracy = 0.8571
Class 1: Accuracy = 1.0000



Classification Report:

	precision	recall	f1-score	support
0	0.83	0.91	0.87	11
1	0.93	0.87	0.90	15
accuracy			0.88	26
macro avg	0.88	0.89	0.88	26
weighted avg	0.89	0.88	0.89	26

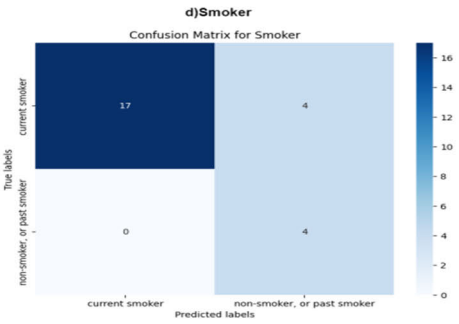
Class 0: Accuracy = 0.9091
Class 1: Accuracy = 0.8667



Classification Report:

	precision	recall	f1-score	support
0	1.00	0.89	0.94	19
1	0.75	1.00	0.86	6
accuracy			0.92	25
macro avg	0.88	0.95	0.90	25
weighted avg	0.94	0.92	0.92	25

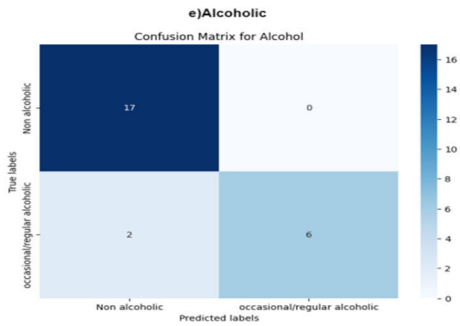
Class 0: Accuracy = 0.8947
Class 1: Accuracy = 1.0000



Classification Report:

	precision	recall	f1-score	support
0	1.00	0.81	0.89	21
1	0.50	1.00	0.67	4
accuracy			0.84	25
macro avg	0.75	0.90	0.78	25
weighted avg	0.92	0.84	0.86	25

Class 0: Accuracy = 0.8095
Class 1: Accuracy = 1.0000



Classification Report:

	precision	recall	f1-score	support
0	0.89	1.00	0.94	17
1	1.00	0.75	0.86	8
accuracy			0.92	25
macro avg	0.95	0.88	0.90	25
weighted avg	0.93	0.92	0.92	25

Class 0: Accuracy = 1.0000
Class 1: Accuracy = 0.7500

FIGURE 9. Confusion matrix and its classification report for all Risk factors.

TABLE 11. Comparison comparative analysis using Resnet 50 and a method without Resnet 50.

Model	Accuracy
With ResNet 50	0.91
Without ResNet 50	0.72
CNN	0.71
Without CNN	0.69
With VGG-19	0.81
Without VGG-19	0.70

Finally, a novel approach has been introduced that utilizes retinal images to quantitatively analyze various cardiovascular risk factors, including gender, age, MI, alcohol intake, and smoking. This evaluation method incorporates the use of ResNet50. The precise evaluation of risk variables in retinal fundus photos has substantial significance in the diagnosis process of CVD, hence we did a comparative assessment with and without ResNet50.The outcomes of this comparison in Table 11.

VI. CONCLUSION & FUTURE WORK

This study reinforces the clinical relevance of analyzing the microvascular anatomy and function of the retina as a non-invasive means to assess systemic CVD risk. We proposed a advanced driven diagnostic methodology that incorporates retinal vasculature imaging specifically the measurement of AVR and CDR alongside established systemic biomarkers such as age, sex, BMI, smoking, and alcohol status. The integration of these retinal and systemic features enables the early identification of individuals at high risk of developing CVD, particularly those already affected by DR or HR. The findings demonstrate that the presence of DR and HR is significantly associated with elevated cardiovascular risk, validating the utility of retinal imaging as a complementary tool in preventive CVD. Overall, this research presents an advanced, non-invasive framework that enhances early diagnosis and clinical decision making, with the potential to reduce CVD incidence through timely intervention.

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Not Applicable.

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